Linear Algebra I Recitation Class: Week 7

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively. V, W are vector spaces over a field F. If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind. If T is a linear transformation, we use the notation $\ker T := N(T)$ and $\operatorname{im} T := R(T)$.

Recall that a linear transformation $T : V \to W$ is an isomorphism if T is 1-1 and onto, equivalently, $\ker T = \{0\}$ and $\operatorname{im} T = W$. In this case, we say V and W are isomorphic and denote this by $V \cong W$. This means that V and W are "essentially the same" after identifying the elements in V as elements of W via T.

We say the diagram

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle j} \\ X & \xrightarrow[h]{} & Y \end{array}$$

commutes if $j \circ f = h \circ g$.

Fix an ordered basis β for V and γ for W; we then have a commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ {\scriptstyle \phi_\beta}\downarrow\wr & & \wr\downarrow{\scriptstyle \phi_\gamma} \\ F^n & \xrightarrow[L_A, A=[T]_\beta^\gamma]{} & F^m \end{array}$$

Here $\phi_\beta(v) := [v]_\beta$ and $\phi_\gamma(w) := [w]_\gamma$. This justifies the matrix representation of linear transformations.

In these notes, we investigate an important structural result of vector spaces and linear transformations called the First Isomorphism Theorem, alongside its corollaries. All of these are useful tools for identifying one specific space with another.

Theorem 1 (First Isomorphism Theorem)**.** *Let V, W be vector spaces and $T : V \to W$ be linear, then we have an isomorphism*

$$\Phi : V/\ker T \xrightarrow{\sim} \operatorname{im} T \qquad v + \ker T \longmapsto T(v).$$

Proof. To save chalk, we write $\overline{v} := v + \ker T$ for any $v \in V$.

Φ is well-defined: If $\overline{v} = \overline{v'}$, then $v - v' \in \ker T$. Thus, $T(v) - T(v') = T(v - v') = 0$, which means $T(v) = T(v')$.

Φ is linear: Given $\overline{v}, \overline{v'} \in V/\ker T$ and $a \in F$, we have

$$\begin{aligned} \Phi(\overline{v} + a\overline{v'}) &= \Phi(\overline{v + av'}) \\ &= T(v + av') \\ &= T(v) + aT(v') = \Phi(\overline{v}) + a\Phi(\overline{v'}). \end{aligned}$$

Φ is one-to-one: If $\overline{v} \in \ker \Phi$, then $\Phi(\overline{v}) = T(v) = 0$, thus $v \in \ker T$, i.e., $\overline{v} = 0_{V/\ker T}$. Thus $\ker \Phi \subseteq \{0_{V/\ker T}\}$ and naturally we have $0_{V/\ker T} \in \ker \Phi$.

Φ is onto: Given $x \in \operatorname{im} T$, $x = T(y)$ for some $y \in V$, which implies $\Phi(\overline{y}) = x$ for $\overline{y} \in V/\ker T$. □

Theorem 2 (Second Isomorphism Theorem)**.** *Let V be a vector space and W, U be subspaces of V. Then*

$$(W+U)/U \cong W/(W \cap U)$$

Proof. Define

$$T : W \longrightarrow (W+U)/U \qquad w \longmapsto w+U =: \overline{w}.$$

Given $w, w' \in W$ and $a \in F$,

$$T(w+aw') = \overline{w+aw'} = \overline{w} + a\overline{w'} = T(w) + aT(w').$$

Hence, T is linear. Given $x \in (W+U)/U$, we can write $x = \overline{w+u} = \overline{w}$ for some $w \in W$ and $u \in U$. Therefore, $T(w) = \overline{w} = x$, meaning T is onto.

We claim that $\ker T = W \cap U$. If $u \in W \cap U$, then $T(u) = \overline{u} = \overline{0}$ since $u \in U$, so $W \cap U \subseteq \ker T$. On the other hand, given $w \in \ker T \subseteq W$, we have $T(w) = \overline{w} = \overline{0}$, therefore $w \in U$. So $w \in W \cap U$, and thus $\ker T = W \cap U$.

By the First Isomorphism Theorem, we have

$$W/\ker T \cong \operatorname{im} T \implies W/(W \cap U) \cong (W+U)/U.$$

□

Theorem 3 (Third Isomorphism Theorem)**.** *Let V be a vector space, and let $W \subseteq W_1 \subseteq W_2$ be subspaces of V. Then W_1/W is a subspace of W_2/W and*

$$(W_2/W)/(W_1/W) \cong W_2/W_1.$$

Proof. Define

$$T : W_2/W \longrightarrow W_2/W_1 \qquad w+W \longmapsto w+W_1$$

T is well-defined: if $w+W = w'+W$, then $w-w' \in W \subseteq W_1$, thus $w+W_1 = w'+W_1$. The linearity of T follows directly from the vector addition and scalar multiplication of quotient spaces. T is onto since for any $x = w+W_1 \in W_2/W_1$, we have $w \in W_2$, and $T(w+W) = x$ with $w+W \in W_2/W$.

We claim that $\ker T = W_1/W$. If $x = w+W \in W_1/W$, then $T(x) = w+W_1 = 0+W_1$ since $w \in W_1$, so $W_1/W \subseteq \ker T$. On the other hand, for $x = w+W \in \ker T$, $T(x) = w+W_1 = 0+W_1$, thus $w \in W_1$, i.e., $x \in W_1/W$. Consequently, $\ker T = W_1/W$, and W_1/W is a subspace of W_2/W.

Hence, by the First Isomorphism Theorem, we have

$$(W_2/W)/\ker T \cong \operatorname{im} T \implies (W_2/W)/(W_1/W) \cong W_2/W_1.$$

□

The following theorem is sometimes called the Fourth Isomorphism Theorem, even though it has nothing to do with isomorphism. The alternative name for this theorem is considered more descriptive of its statement.

Theorem 4 (Correspondence Theorem)**.** *Let V be a vector space and W be a subspace of V. We have a 1-1 correspondence (i.e., a bijection between sets):*

$$\left\{ \begin{array}{c} \textit{Vector subspaces } U \textit{ of } V \\ \textit{such that } W \subseteq U \end{array} \right\} \longleftrightarrow \left\{ \textit{ Vector subspaces of } V/W \right\}$$
$$U \longmapsto \pi(U)$$
$$\pi^{-1}(Z) \longmapsto\!\shortmid\ Z$$

where $\pi : V \longrightarrow V/W, v \longmapsto v + W$ is the canonical projection.

Proof. Recall that the image (resp. inverse image) of a vector subspace is a vector subspace of the codomain (resp. domain). Furthermore, given any vector subspace Z of V/W, since $0_{V/W} \in Z$, it follows that $W = \ker \pi = \pi^{-1}(0) \subseteq \pi^{-1}(Z)$. Therefore, the maps from one side to another are both well-defined (here by well-defined, we mean the image of the function truly lies in the specified codomain). To prove the 1-1 correspondence, it suffices to show that:

1. $\pi^{-1}(\pi(U)) = U$ for any vector subspace U of V such that $W \subseteq U$.
2. $\pi(\pi^{-1}(Z)) = Z$ for any vector subspace Z of V/W.

For the first statement, given $x \in \pi^{-1}(\pi(U))$, we have $\pi(x) \in \pi(U) = U/W$. Thus, $\pi(x) = u + W$ for some $u \in U$, meaning $x - u \in W$. Since $x - u = w$ for some $w \in W$, we have $x = u + w \in U$ (because $W \subseteq U$). On the other hand, for any $u \in U$, $\pi(u) \in \pi(U)$ so $u \in \pi^{-1}(\pi(U))$.

A similar argument applies to prove the second statement. Given $z \in Z$, since π is surjective, there exists $v \in V$ such that $\pi(v) = z$. Thus $v \in \pi^{-1}(Z)$, meaning $z = \pi(v) \in \pi(\pi^{-1}(Z))$. Conversely, it is clear from the definition of the inverse image that $\pi(\pi^{-1}(Z)) \subseteq Z$. □

Remark. The isomorphism theorems hold for various kinds of algebraic structures such as groups (with group homomorphisms), rings (with ring homomorphisms), and modules (with module homomorphisms).

Example 5. Let $V = M_n(\mathbb{R})$ and define $W = \{A \in V \mid \operatorname{tr}(A) = 0\}$. One can check that W is a subgroup of V. We will use the first isomorphism theorem to find the dimension of W. Consider the linear map

$$\phi : V \longrightarrow \mathbb{R} \qquad A \longmapsto \operatorname{tr}(A).$$

ϕ is onto since for any $c \in \mathbb{R}$, define $B = cE_{11}$, where E_{ij} is the matrix with 1 at the ij-entry and 0 at all the others, then $\phi(B) = c$. Note that W is defined as the set with matrices A such that $\phi(A) = \operatorname{tr}(A) = 0$, therefore, $\ker \phi = W$. By the first isomorphism theorem, we have

$$V/\ker \phi \cong \operatorname{im} \phi \implies V/W \cong \mathbb{R}.$$

Hence, $1 = \dim \mathbb{R} = \dim(V/W) = \dim V - \dim W = n^2 - \dim W$ since V is finite dimensional. $\dim W = n^2 - 1$.

Remark. From the example above, we know that it is not necessary to find a basis to compute the dimension of some vector spaces.

Exercise A. Find a basis for $W = \{A \in M_n(\mathbb{R}) \mid \operatorname{tr}(A) = 0\}$. It should be a set with $n^2 - 1$ elements.

References

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